A CLASSICAL GENERATOR IS A GENERATOR

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11276-11283, immediately following the statement at 11269-11274.

Editorial note. The id_K = 0 reading is the span's own closing step.

Allowed standard prerequisites (editorial).

- Right orthogonality extends to the envelope. Lemma lemma-right-orthogonal of this chapter, cited at proof lines 11279-11280. Vanishing against E in all degrees extends to vanishing against every object of the envelope.

Source reference. Original lemma 09SK: sources/stacks-derived/pinned-original-source.tex, source lines 11251–11251. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a triangulated category. Let E be an object of $\mathcal{D}$. If E is a classical generator of $\mathcal{D}$, then E is a generator.*

Proof. Assume E is a classical generator. Let K be an object of $\mathcal{D}$ such that $\mathrm{Hom}(E, K[i]) = 0$ for all $i \in \mathbf{Z}$. By Lemma 09SK $\mathrm{Hom}(E', K) = 0$ for all E' in $\langle E \rangle$. However, since $\mathcal{D} = \langle E \rangle$ we conclude that $\mathrm{id}_K = 0$, i.e., $K = 0$. □

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